

SECTION 4 METHODS OF STRUCTURAL ANALYSIS

4.1 Methods of Determining Action Effects

4.1.1 General

For the purpose of complying with the requirements of the limit states of stability, strength and serviceability specified in Section 5, effects of design actions on a structure and its members and connections, shall be determined by structural analysis using the assumptions of 4.2 and 4.3 and one of the following methods of analysis:

- a) Elastic analysis in accordance with 4.4,
- b) Plastic analysis in accordance with 4.5,
- c) Advanced analysis in accordance with Annex B, and
- d) Dynamic analysis in accordance with IS 1893 (Part 1).

The design action effects for design basis earthquake loads shall be obtained only by an elastic analysis. The maximum credible earthquake loads shall be assumed to correspond to the load at which significant plastic hinges are formed in the structure and the corresponding effects shall be obtained by plastic or advanced analysis. More information on analysis and design to resist earthquake is given in Section 12 and IS 1893 (Part 1).

4.1.2 Non-sway and Sway Frames

For the purpose of analysis and design, the structural frames are classified as non-sway and sway frames as given below:

- a) *Non-sway frame* — One in which the transverse displacement of one end of the member relative to the other end is effectively prevented. This applies to triangulated frames and trusses or to frames where in-plane stiffness is provided by bracings, or by shear walls, or by floor slabs and roof decks secured horizontally to walls or to bracing systems parallel to the plane of loading and bending of the frame.
- b) *Sway frame* — One in which the transverse displacement of one end of the member relative to the other end is not effectively prevented. Such members and frames occur in structures which depend on flexural action of members to resist lateral loads and sway, as in moment resisting frames.
- c) A rigid jointed multi-storey frame may be considered as a non-sway frame if in every individual storey, the deflection δ , over a storey height h_s , due to the notional horizontal

loading given in 4.3.6 satisfies the following criteria:

- 1) For clad frames, when the stiffening effect of the cladding is not taken into account in the deflection calculations:

$$\delta \leq \frac{h_s}{2\ 000}$$

- 2) For unclad frame or clad frames, when the stiffening effect of the cladding is taken into account in the deflection calculations:

$$\delta \leq \frac{h_s}{4\ 000}$$

- 3) A frame, which when analyzed considering all the lateral supporting system does not comply with the above criteria, should be classified as a sway frame, even if it is braced or otherwise laterally stiffened.

4.2 Forms of Construction Assumed for Structural Analysis

4.2.1 The effects of design action in the members and connections of a structure shall be determined by assuming singly or in combination of the following forms of construction (*see 10.6.1*).

4.2.1.1 Rigid construction

In rigid construction, the connections between members (beam and column) at their junction shall be assumed to have sufficient rigidity to hold the original angles between the members connected at a joint unchanged under loading.

4.2.1.2 Semi-rigid construction

In semi-rigid construction, the connections between members (beam and column) at their junction may not have sufficient rigidity to hold the original angles between the members at a joint unchanged, but shall be assumed to have the capacity to furnish a dependable and known degree of flexural restraint. The relationship between the degree of flexural restraint and the level of the load effects shall be established by any rational method or based on test results (*see Annex F*).

4.2.1.3 Simple construction

In simple construction, the connections between members (beam and column) at their junction will not resist any appreciable moment and shall be assumed to be hinged.

4.2.2 Design of Connections

The design of all connections shall be consistent with

the form of construction, and the behaviour of the connections shall not adversely affect any other part of the structure beyond what is allowed for in design. Connections shall be designed in accordance with Section 10.

4.3 Assumptions in Analysis

4.3.1 The structure shall be analyzed in its entirety except as follows:

- a) Regular building structures, with orthogonal frames in plan, may be analyzed as a series of parallel two-dimensional sub-structures (part of a structure), the analysis being carried out in each of the two directions, at right angles to each other, except when there is significant load redistribution between the sub-structures (part of a structure). For earthquake loading three dimensional analysis may be necessary to account for effects of torsion and also for multi-component earthquake forces [see IS 1893 (Part 1)].
- b) For vertical loading in a multi-storey building structure, provided with bracing or shear walls to resist all lateral forces, each level thereof, together with the columns immediately above and below, may be considered as a sub-structure, the columns being assumed fixed at the ends remote from the level under consideration.
- c) Where beams at a floor level in a multi-bay building structure are considered as a sub-structure (part of a structure), the bending moment at the support of the beam due to gravity loads may be determined based on the assumption that the beam is fixed at the far end support, one span away from the span under consideration, provided that the floor beam is continuous beyond that support point.

4.3.2 Span Length

The span length of a flexural member in a continuous frame system shall be taken as the distance between centre-to-centre of the supports.

4.3.3 Arrangements of Imposed Loads in Buildings

For building structures, the various arrangements of imposed loads considered for the analysis, shall include at least the following:

- a) Where the loading pattern is fixed, the arrangement concerned.
- b) Where the imposed load is variable and not greater than three-quarters of the dead load, the live load may be taken to be acting on all spans.

- c) Where the imposed load is variable and exceeds three-quarters of the dead load, arrangements of live load acting on the floor under consideration shall include the following cases:

- 1) Imposed load on alternate spans,
- 2) Imposed load on two adjacent spans, and
- 3) Imposed load on all the spans.

4.3.4 Base Stiffness

In the analysis of all structures the appropriate base stiffness about the axis under consideration shall be used. In the absence of the knowledge of the pedestal and foundation stiffness, the following may be assumed:

- a) When the column is rigidly connected to a suitable foundation, the stiffness of the pedestal shall be taken as the stiffness of the column above base plate. However in case of very stiff pedestals and foundations the column may be assumed as fixed at base.
- b) When the column is nominally connected to the foundation, a pedestal stiffness of 10 percent of the column stiffness may be assumed.
- c) When an actual pin or rocker is provided in the connection between the steel column and pedestal, the column is assumed as hinged at base and the pedestal and foundation may be appropriately designed for the reactions from the column.
- d) In case of (a) and (b), the bottom of the pedestal shall be assumed to have the following boundary condition in the absence of any detailed procedure based on theory or tests:
 - 1) When the foundation consist of a group of piles with a pile cap, raft foundation or an isolated footing resting on rock or very hard soil, the pedestal shall be assumed to be fixed at the level of the bottom of footing or at the top of pile cap.
 - 2) When the foundation consist of an isolated footing resting on other soils, pedestal shall be assumed to be hinged at the level of the bottom of footing.
 - 3) When the pedestal is supported by a single pile, which is laterally surrounded by soil providing passive resistance, the pile shall be assumed to be fixed at a depth of 5 times the diameter of the pile

below the ground level in case of compact ground or the top level of compact soil in case of poor soil overlying compact soil.

- 4) When the column is founded into rock, it may be assumed to be fixed at the interface of the column and rock.

4.3.5 Simple Construction

Bending members may be assumed to have their ends connected for shear only and to be free to rotate. In triangulated structures, axial forces may be determined by assuming that all members are pin connected. The eccentricity for stanchion and column shall be assumed in accordance with 7.3.3.

4.3.6 Notional Horizontal Loads

To analyze a frame subjected to gravity loads, considering the sway stability of the frame, notional horizontal forces should be applied. These notional horizontal forces account for practical imperfections and should be taken at each level as being equal to 0.5 percent of factored dead load plus vertical imposed loads applied at that level. The notional load should not be applied along with other lateral loads such as wind and earthquake loads in the analysis.

4.3.6.1 The notional forces should be applied on the whole structure, in both orthogonal directions, in one direction at a time, at roof and all floor levels or their equivalent. They should be taken as acting simultaneously with factored gravity loads.

4.3.6.2 The notional force should not be,

- a) applied when considering overturning or overall instability;
- b) combined with other horizontal (lateral) loads;
- c) combined with temperature effects; and
- d) taken to contribute to the net shear on the foundation.

4.3.6.3 The sway effect using notional load under gantry load case need not be considered if the ratio of height to lateral width of the building is less than unity.

4.4 Elastic Analysis

4.4.1 Assumptions

Individual members shall be assumed to remain elastic under the action of the factored design loads for all limit states.

The effect of haunching or any variation of the cross-section along the axis of a member shall be considered, and where significant, shall be taken into account in the determination of the member stiffness.

4.4.2 First-Order Elastic Analysis

In a first-order elastic analysis, the equilibrium of the frame in the undeformed geometry is considered, the changes in the geometry of the frame due to the loading are not accounted for, and changes in the effective stiffness of the members due to axial force are neglected. The effects of these on the first-order bending moments shall be allowed for by using one of the methods of moment amplification of 4.4.3.2 or 4.4.3.3 as appropriate. Where the moment amplification factor C_y , C_z , calculated in accordance with 4.4.3.2 or 4.4.3.3 as appropriate, is greater than 1.4, a second-order elastic analysis in accordance with Annex B shall be carried out.

4.4.3 Second-Order Elastic Analysis

4.4.3.1 The analysis shall allow for the effects of the design loads acting on the structure and its members in their displaced and deformed configuration. These second-order effects shall be taken into account by using either:

- a) A first-order elastic analysis with moment amplification in accordance with 4.4.2, provided the moment amplification factors, C_y and C_z are not greater than 1.4; or
- b) A second-order elastic analysis in accordance with Annex B.

4.4.3.2 Moment amplification for members in non-sway frames

For a member with zero axial compression or a member subject to axial tension, the design bending moment is that obtained from the first order analysis for factored loads, without any amplification.

For a braced member with a design axial compressive force P_d as determined by the first order analysis, the design bending moment shall be calculated considering moment amplification as in 9.3.2.2.

4.4.3.3 Moment amplification for members in sway frames

The design bending moment shall be calculated as the product of moment amplification factor [see 9.3.2.2 (C_{my} , C_{mz})] and the moment obtained from the first order analysis of the sway frame, unless analysis considering second order effects is carried out (see 4.4.3).

4.4.3.4 The calculated bending moments from the first order elastic analysis may be modified by redistribution upto 15 percent of the peak calculated moment of the member under factored load, provided that:

- a) the internal forces and moments in the

members of the frame are in equilibrium with applied loads.

- b) all the members in which the moments are reduced shall belong to plastic or compact section classification (*see 3.7*).

4.5 Plastic Analysis

4.5.1 Application

The effects of design action throughout or on part of a structure may be determined by a plastic analysis, provided that the requirements of 4.5.2 are met. The distribution of design action effects shall satisfy equilibrium and the boundary conditions.

4.5.2 Requirements

When a plastic method of analysis is used, all of the following conditions shall be satisfied, unless adequate ductility of the structure and plastic rotation capacity of its members and connections are established for the design loading conditions by other means of evaluation:

- a) The yield stress of the grade of the steel used shall not exceed 450 MPa.
- b) The stress-strain characteristics of the steel shall not be significantly different from those obtained for steels complying with IS 2062 or equivalent and shall be such as to ensure complete plastic moment redistribution. The stress-strain diagram shall have a plateau at the yield stress, extending for at least six times the yield strain. The ratio of the tensile strength to the yield stress specified for the grade of the steel shall not be less than 1.2. The elongation on a gauge length complying with IS 2062 shall not be than 15 percent, and the steel shall exhibit strain-hardening capability. Steels conforming to IS 2062 shall be deemed to satisfy the above requirements.
- c) The members used shall be hot-rolled or fabricated using hot-rolled plates and sections.
- d) The cross-section of members not containing plastic hinges should be at least that of compact section (*see 3.7.2*), unless the members meet the strength requirements from elastic analysis.
- e) Where plastic hinges occur in a member, the proportions of its cross-section should not exceed the limiting values for plastic section given in 3.7.2.
- f) The cross-section should be symmetrical about its axis perpendicular to the axis of the plastic hinge rotation.
- g) The members shall not be subject to impact loading, requiring fracture assessment or

fluctuating loading, requiring a fatigue assessment (*see Section 13*).

4.5.2.1 Restraints

If practicable, torsional restraint (against lateral buckling) should be provided at all plastic hinge locations. Where not feasible, the restraint should be provided within a distance of $D/2$ of the plastic hinge location, where D is the total depth of section.

The torsional restraint requirement at a section as above, need not be met at the last plastic hinge to form, provided it can be clearly identified.

Within a member containing a plastic hinge, the maximum distance L_m from the restraint at the plastic hinge to an adjacent restraint should be calculated by any rational method or the conservative method given below, so as to prevent lateral buckling.

Conservatively L_m (in mm) may be taken as

$$L_m \leq \frac{38 r_y}{\left[\frac{f_c}{130} + \left(\frac{f_y}{250} \right)^2 \left(\frac{x_t}{40} \right)^2 \right]^{1/2}}$$

where

f_c = actual compressive stress on the cross-section due to axial load, in N/mm²;

f_y = yield stress, in N/mm²;

r_y = radius of gyration about the minor axis, in mm;

x_t = torsional index, $x_t = 1.132 (AI_w/I_y I_t)^{0.5}$;

A = area of cross-section; and

I_w, I_y, I_t = warping constant, second moment of the cross section about the minor axes and St. Venant's torsion constant, respectively.

Where the member has unequal flanges, r_y should be taken as the lesser of the values of the compression flange only or the whole section.

Where the cross-section of the member varies within the length L_m , the maximum value of r_y and the minimum value of x_t should be used.

The spacing of restraints to member lengths not containing a plastic hinge should satisfy the recommendations of section on lateral buckling strength of beams (*see 8.2.2*). Where the restraints are placed at the limiting distance L_m , no further checks are required.

4.5.2.2 Stiffeners at plastic hinge locations

Web stiffeners should be provided where a concentrated load, which exceeds 10 percent of the shear capacity

of the member, is applied within $D/2$ of a plastic hinge location (see 8.2.1.2). The stiffener should be provided within a distance of half the depth of the member on either side of the hinge location and be designed to carry the applied load in accordance with 8.7.4. If the stiffeners are flat plates, the outstand width to the thickness ratio, b/t , should not exceed the values given in the plastic section (see 3.7, Table 2). Where other

sections are used the ratio $\left(\frac{I_{so}}{I_t}\right)^{1/2}$ should not exceed

the values given for plastic section (for simple outstand, as in 3.7);

where

I_{so} = second moment of area of the stiffener about the face of the element perpendicular to the web; and

I_t = St. Venant's torsion constant of the stiffener.

4.5.2.3 The frame shall be adequately supported against sway and out-of-plane buckling, by bracings, moment resisting frame or an independent system such as shear wall.

4.5.2.4 Fabrication restriction

Within a length equal to the member depth, on either side of a plastic hinge location, the following restrictions should be applied to the tension flange and noted in the design drawings. Holes if required, should be drilled or else punched 2 mm undersize and reamed. All sheared or hand flame cut edges should be finished smooth by grinding, chipping or planning.

4.5.3 Assumptions in Analysis

The design action effects shall be determined using a rigid-plastic analysis.

It shall be permissible to assume full strength or partial strength connections, provided the capacities of these are used in the analysis, and provided that

- in a full strength connection, the moment capacity of the connection shall be not less than that of the member being connected;
- in a partial strength connection, the moment capacity of the connection may be less than that of the member being connected; and
- in both cases the behaviour of the connection shall be such as to allow all plastic hinges necessary for the collapse mechanism to develop, and shall be such that the required plastic hinge rotation does not exceed the rotation capacity at any of the plastic hinges in the collapse mechanism.

In the case of building structures, it is not normally necessary to consider the effect of alternating plasticity.

4.5.4 Second-Order Elastic Analysis

Any second-order effects of the loads acting on the structure in its deformed configuration may be neglected, provided the following are satisfied:

- For clad frames, provided the stiffening effects of masonry infill wall panels or diaphragms of profiled wall panel is not taken into account, and where elastic buckling load factor, λ_{cr} (see 4.6) satisfies $\lambda_{cr}/\lambda_p \geq 10$.
If $10 > \lambda_{cr}/\lambda_p \geq 4.6$ the second-order effects may be considered by amplifying the design load effects obtained from plastic analysis by a factor $\delta_p = \{0.9 \lambda_{cr} / (\lambda_{cr} - 1)\}$.
If $\lambda_{cr} / \lambda_p < 4.6$, second-order elasto-plastic analysis or second-order elastic analysis (see 4.4.3) is to be carried out.
- For un-clad frames or for clad frames where the stiffening effects of masonry infill or diaphragms of profiled wall panel is taken into account, where elastic buckling load factor, λ_{cr} (see 4.6) satisfies $\lambda_{cr} / \lambda_p \geq 20$.
If $20 > \lambda_{cr} / \lambda_p \geq 5.75$ the second-order effects may be considered by amplifying the design load effects obtained from plastic analysis by a factor $\delta_p = \{0.9 \lambda_{cr} / (\lambda_{cr} - 1)\}$.
If $\lambda_{cr} / \lambda_p < 5.75$, second-order elasto-plastic analysis or second-order elastic analysis (see 4.4.3) shall be carried out.

4.6 Frame Buckling Analysis

4.6.1 The elastic buckling load factor (λ_{cr}) shall be the ratio of the elastic buckling load set of the frame to the design load set for the frame, and shall be determined in accordance with 4.6.2.

NOTE — The value of λ_{cr} depends on the load set and has to be evaluated for each possible set of load combination.

4.6.2 In-plane Frame Buckling

The elastic buckling load factor (λ_{cr}) of a rigid-jointed frame shall be determined by using:

- One of the approximate methods of 4.6.2.1 and 4.6.2.2, or
- A rational elastic buckling analysis of the whole frame.

4.6.2.1 Regular non-sway frames (see 4.1.2)

In a rectangular non-sway frame with regular loading and negligible axial forces in the beams, the Euler buckling stress f_{cc} for each column shall be determined in accordance with 7.1.2.1. The elastic buckling load factor (λ_{cr}) for the whole frame shall be taken as the lowest of the ratio of (f_{cc}/f_{cd}) for all the columns, where

f_{cd} is the axial compressive stress in the column from the factored load analysis.

4.6.2.2 Regular sway frames

In a rectangular sway frame with regular loading and negligible axial forces in the beams, the buckling load, P_{cc} , for each column shall be determined as $P_{cc} = A f_{cc}$ where f_{cc} is the elastic buckling stress of the column in the plane of frame, obtained in accordance with 7.1.2.1. The elastic buckling load factor λ_{scr} for the whole frame shall be taken as the lowest of all the ratios, λ_{scr} , calculated for each storey of the building, as given below:

$$\lambda_{scr} = \frac{\sum (P_{cc} / L)}{\sum (P / L)}$$

where

- P = member axial force from the factored load analysis, with tension taken as negative; and
- L = column length and the summation includes all columns in the plane frame within a storey.

SECTION 5 LIMIT STATE DESIGN

5.1 Basis for Design

5.1.1 In the limit state design method, the structure shall be designed to withstand safely all loads likely to act on it throughout its life. It shall not suffer total collapse under accidental loads such as from explosions or impact or due to consequences of human error to an extent beyond the local damages. The objective of the design is to achieve a structure that will remain fit for use during its life with acceptable target reliability. In other words, the probability of a limit state being reached during its lifetime should be very low. The acceptable limit for the safety and serviceability requirements before failure occurs is called a limit state. In general, the structure shall be designed on the basis of the most critical limit state and shall be checked for other limit states.

5.1.2 Steel structures are to be designed and constructed to satisfy the design requirements with regard to stability, strength, serviceability, brittle fracture, fatigue, fire, and durability such that they meet the following:

- a) Remain fit with adequate reliability and be able to sustain all actions (loads) and other influences experienced during construction and use;
- b) Have adequate durability under normal maintenance;
- c) Do not suffer overall damage or collapse

disproportionately under accidental events like explosions, vehicle impact or due to consequences of human error to an extent beyond local damage. The potential for catastrophic damage shall be limited or avoided by appropriate choice of one or more of the following:

- 1) Avoiding, eliminating or reducing exposure to hazards, which the structure is likely to sustain.
- 2) Choosing structural forms, layouts and details and designing such that:
 - i) the structure has low sensitivity to hazardous conditions; and
 - ii) the structure survives with only local damage even after serious damage to any one individual element by the hazard.
- 3) Choosing suitable material, design and detailing procedure, construction specifications, and control procedures for shop fabrication and field construction as relevant to the particular structure.

The following conditions may be satisfied to avoid a disproportionate collapse:

- a) The building should be effectively tied together at each principal floor level and each column should be effectively held in position by means of continuous ties (beams) nearly orthogonal, except where the steel work supports only cladding weighing not more than 0.7 kN/m² along with imposed and wind loads. These ties must be steel members such as beams, which may be designed for other purposes, steel bar reinforcement anchoring the steel frame to concrete floor or steel mesh reinforcement in composite slab with steel profiled sheeting directly connected to beam with shear connectors. These steel ties and their end connections should be capable of resisting factored tensile force not less than the factored dead and imposed loads acting on the floor area tributary to the tie nor less than 75 kN. Such connection of ties to edge column should also be capable of resisting 1 percent of the maximum axial compression in the column at the level due to factored dead and imposed loads. All column splices should be capable of resisting a tensile force equal to the largest of a factored dead and live load reaction from a single floor level located between that column splice and the next column splice below that splice. Lateral load system to resist notional horizontal loads prescribed in 4.3.6 should be distributed

throughout the building in nearly orthogonal directions so that no substantial portions is connected at only one point to such a system. Precast concrete or other heavy floor or roof units should be effectively anchored in the direction of their span either to each other over the support or directly to the support.

- b) Where the above conditions to tie the columns to the floor adequately are not satisfied each storey of the building should be checked to ensure that disproportionate collapse would not precipitate by the notional removal, one at a time, of each column.
- c) Where each floor is not laterally supported by more than one system, check should be made at each storey by removing one such lateral support system at a time to ensure that disproportionate collapse would not occur. The collapse is considered disproportionate, if more than 15 percent of the floor or roof area of 70 m² collapse at that level and at one adjoining level either above or below it, under a load equal to 1.05 or 0.9 times the dead load, 0.33 times temporary or full imposed load of permanent nature (as in storage buildings) and 0.33 times wind load acting together.

5.1.3 Structures designed for unusual or special functions shall comply with any other relevant additional limit state considered appropriate to that structure.

5.1.4 Generally structures and elements shall be designed by limit state method. Where limit state method cannot be conveniently adopted, working stress design (*see* Section 11) may be used.

5.2 Limit State Design

5.2.1 For achieving the design objectives, the design shall be based on characteristic values for material strengths and applied loads (actions), which take into account the probability of variations in the material strengths and in the loads to be supported. The characteristic values shall be based on statistical data, if available. Where such data is not available, these shall be based on experience. The design values are derived from the characteristic values through the use of partial safety factors, both for material strengths and for loads. In the absence of special considerations, these factors shall have the values given in this section according to the material, the type of load and the limit state being considered. The reliability of design is ensured by satisfying the requirement:

$$\text{Design action} \leq \text{Design strength}$$

5.2.2 Limit states are the states beyond which the structure

no longer satisfies the performance requirements specified. The limit states are classified as:

- a) Limit state of strength; and
- b) Limit state of serviceability.

5.2.2.1 The limit states of strength are those associated with failures (or imminent failure), under the action of probable and most unfavourable combination of loads on the structure using the appropriate partial safety factors, which may endanger the safety of life and property. The limit state of strength includes:

- a) Loss of equilibrium of the structure as a whole or any of its parts or components.
- b) Loss of stability of the structure (including the effect of sway where appropriate and overturning) or any of its parts including supports and foundations.
- c) Failure by excessive deformation, rupture of the structure or any of its parts or components.
- d) Fracture due to fatigue.
- e) Brittle fracture.

5.2.2.2 The limit state of serviceability include:

- a) Deformation and deflections, which may adversely affect the appearance or effective use of the structure or may cause improper functioning of equipment or services or may cause damages to finishes and non-structural members.
- b) Vibrations in the structure or any of its components causing discomfort to people, damages to the structure, its contents or which may limit its functional effectiveness. Special consideration shall be given to systems susceptible to vibration, such as large open floor areas free of partitions to ensure that such vibrations are acceptable for the intended use and occupancy (*see* Annex C).
- c) Repairable damage or crack due to fatigue.
- d) Corrosion, durability.
- e) Fire.

5.3 Actions

The actions (loads) to be considered in design include direct actions (loads) experienced by the structure due to self weight, external actions etc., and imposed deformations such as that due to temperature and settlements.

5.3.1 Classification of Actions

Actions are classified by their variation with time as given below:

- a) *Permanent actions* (Q_p): Actions due to self-weight of structural and non-structural

components, fittings, ancillaries, and fixed equipment, etc.

- b) *Variable actions* (Q_v): Actions due to construction and service stage loads such as imposed (live) loads (crane loads, snow loads, etc.), wind loads, and earthquake loads, etc.
- c) *Accidental actions* (Q_a): Actions expected due to explosions, and impact of vehicles, etc.

5.3.2 Characteristic Actions (Loads)

5.3.2.1 The Characteristic Actions, Q_c , are the values of the different actions that are not expected to be exceeded with more than 5 percent probability, during the life of the structure and they are taken as:

- a) the self-weight, in most cases calculated on the basis of nominal dimensions and unit weights [see IS 875 (Part 1)].
- b) the variable loads, values of which are specified in relevant standard [see IS 875 (all Parts) and IS 1893 (Part 1)].
- c) the upper limit with a specified probability (usually 5 percent) not exceeding during some reference period (design life).
- d) specified by client, or by designer in consultation with client, provided they satisfy the minimum provisions of the relevant loading standard.

5.3.2.2 The characteristic values of accidental loads generally correspond to the value specified by relevant code, standard or client. The design for accidental load

is generally not required in building unless it is required by client or approving authority in which case, generally recommendation in 5.1.2 c) or specialist literature shall be followed.

5.3.3 Design Actions

The Design Actions, Q_d , is expressed as $Q_d = \sum_k \gamma_{fk} Q_{ck}$

where

γ_{fk} = partial safety factor for different loads k , given in Table 4 to account for:

- a) Possibility of unfavourable deviation of the load from the characteristic value,
- b) Possibility of inaccurate assessment of the load,
- c) Uncertainty in the assessment of effects of the load, and
- d) Uncertainty in the assessment of the limit states being considered.

The loads or load effects shall be multiplied by the relevant γ_f factors, given in Table 4, to get the design loads or design load effects.

5.4 Strength

The ultimate strength calculation may require consideration of the following:

- a) Loss of equilibrium of the structure or any part of it, considered as a rigid body; and
- b) Failure by excessive deformation, rupture or

Table 4 Partial Safety Factors for Loads, γ_f , for Limit States
(Clauses 3.5.1 and 5.3.3)

Combination	Limit State of Strength					Limit State of Serviceability			
	DL	LL ¹⁾		WL/EL	AL	DL	LL ¹⁾		WL/EL
		Leading	Accompanying				Leading	Accompanying	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
DL+LL+CL	1.5	1.5	1.05	—	—	1.0	1.0	1.0	—
DL+LL+CL+ WL/EL	1.2	1.2	1.05	0.6	—	1.0	0.8	0.8	0.8
DL+WL/EL	1.2	1.2	0.53	1.2	—	1.0	—	—	1.0
DL+ER	1.2	1.2	—	—	—	—	—	—	—
DL+LL+AL	(0.9) ²⁾ 1.0	0.35	0.35	—	1.0	—	—	—	—

¹⁾ When action of different live loads is simultaneously considered, the leading live load shall be considered to be the one causing the higher load effects in the member/section.

²⁾ This value is to be considered when the dead load contributes to stability against overturning is critical or the dead load causes reduction in stress due to other loads.

Abbreviations:
DL = Dead load, LL = Imposed load (Live loads), WL = Wind load, CL = Crane load (Vertical/Horizontal), AL = Accidental load, ER = Erection load, EL = Earthquake load.

NOTE — The effects of actions (loads) in terms of stresses or stress resultants may be obtained from an appropriate method of analysis as in 4.

loss of stability of the structure or any part of it including support and foundation.

5.4.1 Design Strength

The Design Strength, S_d , is obtained as given below from ultimate strength, S_u and partial safety factors for materials, γ_m given in Table 5.

$$S_d = S_u / \gamma_m$$

where partial safety factor for materials, γ_m account for:

- Possibility of unfavourable deviation of material strength from the characteristic value,
- Possibility of unfavourable variation of member sizes,
- Possibility of unfavourable reduction in member strength due to fabrication and tolerances, and
- Uncertainty in the calculation of strength of the members.

5.5 Factors Governing the Ultimate Strength

5.5.1 Stability

Stability shall be ensured for the structure as a whole and for each of its elements. This should include, overall frame stability against overturning and sway, as given in 5.5.1.1 and 5.5.1.2.

5.5.1.1 Stability against overturning

The structure as a whole or any part of it shall be designed to prevent instability due to overturning, uplift or sliding under factored load as given below:

- The Actions shall be divided into components aiding instability and components resisting instability.
- The permanent and variable actions and their effects causing instability shall be combined using appropriate load factors as per the Limit State requirements, to obtain maximum destabilizing effect.

- The permanent actions (loads) and effects contributing to resistance shall be multiplied with a partial safety factor 0.9 and added together with design resistance (after multiplying with appropriate partial safety factor). Variable actions and their effects contributing to resistance shall be disregarded.
- The resistance effect shall be greater than or equal to the destabilizing effect. Combination of imposed and dead loads should be such as to cause most severe effect on overall stability.

5.5.1.2 Sway stability

The whole structure, including portions between expansion joints, shall be adequately stiff against sway. To ensure this, in addition to designing for applied horizontal loads, a separate check should be carried out for notional horizontal loads such as given in 4.3.6 to evaluate the sway under gravity loads.

5.5.2 Fatigue

Generally fatigue need not be considered unless a structure or element is subjected to numerous significant fluctuations of stress. Stress changes due to fluctuations in wind loading normally need not be considered. Fatigue design shall be in accordance with Section 13. When designing for fatigue, the partial safety factor for load, γ_f equal to unity shall be used for the load causing stress fluctuation and stress range.

5.5.3 Plastic Collapse

Plastic analysis and design may be used, if the requirement specified under the plastic method of analysis (*see* 4.5) are satisfied.

5.6 Limit State of Serviceability

Serviceability limit state is related to the criteria governing normal use. Serviceability limit state is limit state beyond which the service criteria specified below, are no longer met:

- Deflection limit,

Table 5 Partial Safety Factor for Materials, γ_m

(Clause 5.4.1)

Sl No.	Definition	Partial Safety Factor	
i)	Resistance, governed by yielding, γ_{m0}	1.10	
ii)	Resistance of member to buckling, γ_{m0}	1.10	
iii)	Resistance, governed by ultimate stress, γ_{m1}	1.25	
iv)	Resistance of connection:	<i>Shop Fabrications</i>	<i>Field Fabrications</i>
a)	Bolts-Friction Type, γ_{mf}	1.25	1.25
b)	Bolts-Bearing Type, γ_{mb}	1.25	1.25
c)	Rivets, γ_{mr}	1.25	1.25
d)	Welds, γ_{mw}	1.25	1.50

- b) Vibration limit,
- c) Durability consideration, and
- d) Fire resistance.

Unless specified otherwise, partial safety factor for loads, γ_f , of value equal to unity shall be used for all loads leading to serviceability limit states to check the adequacy of the structure under serviceability limit states.

5.6.1 Deflection

The deflection under serviceability loads of a building

or a building component should not impair the strength of the structure or components or cause damage to finishings. Deflections are to be checked for the most adverse but realistic combination of service loads and their arrangement, by elastic analysis, using a load factor of 1.0. Table 6 gives recommended limits of deflections for certain structural members and systems. Circumstances may arise where greater or lesser values would be more appropriate depending upon the nature of material in element to be supported (vulnerable to cracking or not) and intended use of the structure, as required by client.

Table 6 Deflection Limits

Type of Building	Deflection	Design Load	Member	Supporting	Maximum Deflection
(1)	(2)	(3)	(4)	(5)	(6)
Industrial Buildings	Vertical	Live load/ Wind load	Purlins and Girts	Elastic cladding	Span/150
				Brittle cladding	Span/180
		Live load	Simple span	Elastic cladding	Span/240
				Brittle cladding	Span/300
		Live load	Cantilever span	Elastic cladding	Span/120
				Brittle cladding	Span/150
		Live load/ Wind load	Rafter supporting	Profiled Metal Sheeting	Span/180
				Plastered Sheeting	Span/240
	Lateral	Crane load (Manual operation)	Gantry	Crane	Span/500
		Crane load (Electric operation up to 50 t)	Gantry	Crane	Span/750
		Crane load (Electric operation over 50 t)	Gantry	Crane	Span/1 000
		No cranes	Column	Elastic cladding	Height/150
				Masonry/Brittle cladding	Height/240
				Crane (absolute)	Span/400
		Crane + wind	Gantry (lateral)	Relative displacement between rails supporting crane	10 mm
Other Buildings	Vertical	Live load	Floor and Roof	Gantry (Elastic cladding; pendent operated)	Height/200
				Gantry (Brittle cladding; cab operated)	Height/400
		Live load	Cantilever	Elements not susceptible to cracking	Span/300
				Elements susceptible to cracking	Span/360
	Lateral	Wind	Building	Elements not susceptible to cracking	Span/150
				Elements susceptible to cracking	Span/180
		Wind	Inter storey drift	Elastic cladding	Height/300
				Brittle cladding	Height/500
				—	Storey height/300

5.6.1.1 Where the deflection due to the combination of dead load and live load is likely to be excessive, consideration should be given to pre-camber the beams, trusses and girders. The value of desired camber shall be specified in design drawing. Generally, for spans greater than 25 m, a camber approximately equal to the deflection due to dead loads plus half the live load may be used. The deflection of a member shall be calculated without considering the impact factor or dynamic effect of the loads on deflection. Roofs, which are very flexible, shall be designed to withstand any additional load that is likely to occur as a result of ponding of water or accumulation of snow or ice.

5.6.2 Vibration

Suitable provisions in the design shall be made for the dynamic effects of live loads, impact loads and vibration due to machinery operating loads. In severe cases possibility of resonance, fatigue or unacceptable vibrations shall be investigated. Unusually flexible structures (generally the height to effective width of lateral load resistance system exceeding 5:1) shall be investigated for lateral vibration under dynamic wind loads. Structures subjected to large number of cycles of loading shall be designed against fatigue failure, as specified in Section 13. Floor vibration effect shall be considered using specialist literature (*see Annex C*).

5.6.3 Durability

Factors that affect the durability of the buildings, under conditions relevant to their intended life, are listed below:

- Environment,
- Degree of exposure,
- Shape of the member and the structural detail,
- Protective measure, and
- Ease of maintenance.

5.6.3.1 The durability of steel structures shall be ensured by following recommendations in Section 15. Specialist literature may be referred to for more detailed and additional information in design for durability.

5.6.4 Fire Resistance

Fire resistance of a steel member is a function of its mass, its geometry, the actions to which it is subjected, its structural support condition, fire protection measures adopted and the fire to which it is exposed. Design provisions to resist fire are briefly discussed in Section 16. Specialist literature may be referred to for more detailed information in design of fire resistance of steel structures.

SECTION 6 DESIGN OF TENSION MEMBERS

6.1 Tension Members

Tension members are linear members in which axial forces act to cause elongation (stretch). Such members can sustain loads upto the ultimate load, at which stage they may fail by rupture at a critical section. However, if the gross area of the member yields over a major portion of its length before the rupture load is reached, the member may become non-functional due to excessive elongation. Plates and other rolled sections in tension may also fail by block shear of end bolted regions (*see 6.4.1*).

The factored design tension T , in the members shall satisfy the following requirement:

$$T < T_d$$

where

T_d = design strength of the member.

The design strength of a member under axial tension, T_d , is the lowest of the design strength due to yielding of gross section, T_{dg} , rupture strength of critical section, T_{dn} , and block shear T_{db} , given in 6.2, 6.3 and 6.4, respectively.

6.2 Design Strength Due to Yielding of Gross Section

The design strength of members under axial tension, T_{dg} , as governed by yielding of gross section, is given by

$$T_{dg} = A_g f_y / \gamma_{m0}$$

where

f_y = yield stress of the material,

A_g = gross area of cross-section, and

γ_{m0} = partial safety factor for failure in tension by yielding (*see Table 5*).

6.3 Design Strength Due to Rupture of Critical Section

6.3.1 Plates

The design strength in tension of a plate, T_{dn} , as governed by rupture of net cross-sectional area, A_n , at the holes is given by

$$T_{dn} = 0.9 A_n f_u / \gamma_{m1}$$

where

γ_{m1} = partial safety factor for failure at ultimate stress (*see Table 5*),

f_u = ultimate stress of the material, and

A_n = net effective area of the member given by,

$$A_n = \left[b - n d_h + \sum_i \frac{p_{si}^2}{4g_i} \right] t$$

where

b, t = width and thickness of the plate, respectively,

d_h = diameter of the bolt hole (2 mm in addition to the diameter of the hole, in case the directly punched holes),

g = gauge length between the bolt holes, as shown in Fig. 5,

p_s = staggered pitch length between line of bolt holes, as shown in Fig. 5,

n = number of bolt holes in the critical section, and

i = subscript for summation of all the inclined legs.

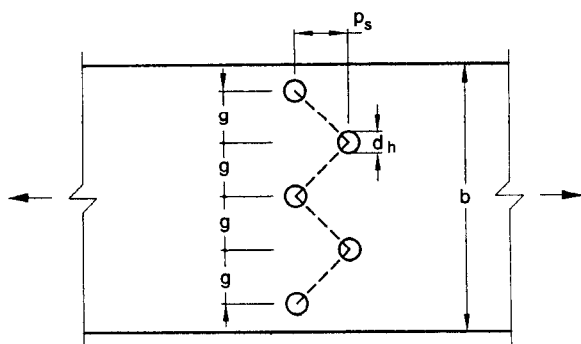


FIG. 5 PLATES WITH BOLTS HOLES IN TENSION

6.3.2 Threaded Rods

The design strength of threaded rods in tension, T_{dn} , as governed by rupture is given by

$$T_{dn} = 0.9 A_n f_u / \gamma_{m1}$$

where

A_n = net root area at the threaded section.

6.3.3 Single Angles

The rupture strength of an angle connected through one leg is affected by shear lag. The design strength, T_{dn} , as governed by rupture at net section is given by:

$$T_{dn} = 0.9 A_{nc} f_u / \gamma_{m1} + \beta A_{go} f_y / \gamma_{m0}$$

where

$$\beta = 1.4 - 0.076 (w/t) (f_y/f_u) (b_s/L_c) \leq (f_u \gamma_{m0} / f_y \gamma_{m1}) \geq 0.7$$

where

w = outstand leg width,

b_s = shear lag width, as shown in Fig. 6, and

L_c = length of the end connection, that is the distance between the outermost bolts in the end joint measured along the load direction or length of the weld along the load direction.

For preliminary sizing, the rupture strength of net section may be approximately taken as:

$$T_{dn} = \alpha A_n f_u / \gamma_{m1}$$

where

α = 0.6 for one or two bolts, 0.7 for three bolts and 0.8 for four or more bolts along the length in the end connection or equivalent weld length;

A_n = net area of the total cross-section;

A_{nc} = net area of the connected leg;

A_{go} = gross area of the outstanding leg; and

t = thickness of the leg.

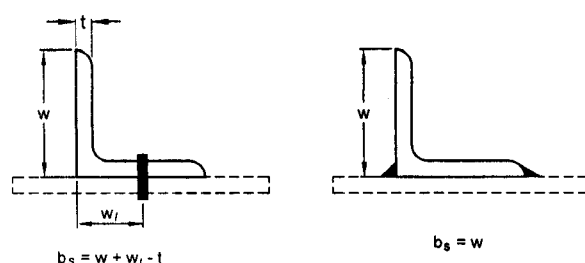


FIG. 6 ANGLES WITH SINGLE LEG CONNECTIONS

6.3.4 Other Section

The rupture strength, T_{dn} , of the double angles, channels, I-sections and other rolled steel sections, connected by one or more elements to an end gusset is also governed by shear lag effects. The design tensile strength of such sections as governed by tearing of net section may also be calculated using equation in 6.3.3, where β is calculated based on the shear lag distance, b_s , taken from the farthest edge of the outstanding leg to the nearest bolt/weld line in the connected leg of the cross-section.

6.4 Design Strength Due to Block Shear

The strength as governed by block shear at an end connection of plates and angles is calculated as given in 6.4.1.

6.4.1 Bolted Connections

The block shear strength, T_{db} of connection shall be taken as the smaller of,

$$T_{db} = [A_{vg} f_y / (\sqrt{3} \gamma_{m0}) + 0.9 A_{tn} f_u / \gamma_{m1}]$$

or

$$T_{db} = (0.9 A_{vn} f_u / (\sqrt{3} \gamma_{m1}) + A_{tg} f_y / \gamma_{m0})$$

where

A_{vg}, A_{vn} = minimum gross and net area in shear along bolt line parallel to external force, respectively (1-2 and 3-4 as shown in Fig. 7A and 1-2 as shown in Fig. 7B),

A_{tg}, A_{tn} = minimum gross and net area in tension from the bolt hole to the toe of the angle, end bolt line, perpendicular to the line of force, respectively (2-3 as shown in Fig. 7B), and

f_u, f_y = ultimate and yield stress of the material, respectively.

6.4.2 Welded Connection

The block shear strength, T_{db} shall be checked for welded end connections by taking an appropriate section in the member around the end weld, which can shear off as a block.

SECTION 7 DESIGN OF COMPRESSION MEMBERS

7.1 Design Strength

7.1.1 Common hot rolled and built-up steel members used for carrying axial compression, usually fail by flexural buckling. The buckling strength of these members is affected by residual stresses, initial bow and accidental eccentricities of load. To account for all these factors, the strength of members subjected to axial compression is defined by buckling class a, b, c, or d as given Table 7.

7.1.2 The design compressive strength P_d , of a member is given by:

$$P < P_d$$

where

$$P_d = A_e f_{cd}$$

where

A_e = effective sectional area as defined in 7.3.2, and

f_{cd} = design compressive stress, obtained as per 7.1.2.1.

7.1.2.1 The design compressive stress, f_{cd} , of axially loaded compression members shall be calculated using the following equation:

$$f_{cd} = \frac{f_y / \gamma_{m0}}{\phi + [\phi^2 - \lambda^2]^{0.5}} = \chi f_y / \gamma_{m0} \leq f_y / \gamma_{m0}$$

where

$$\phi = 0.5 [1 + \alpha (\lambda - 0.2) + \lambda^2]$$

λ = non-dimensional effective slenderness ratio

$$= \sqrt{f_y / f_{cc}} = \sqrt{f_y \left(\frac{KL}{r} \right)^2 / \pi^2 E}$$

$$f_{cc} = \text{Euler buckling stress} = \frac{\pi^2 E}{\left(\frac{KL}{r} \right)^2}$$

where

KL/r = effective slenderness ratio or ratio of effective length, KL to appropriate radius of gyration, r ;

α = imperfection factor given in Table 7;

χ = stress reduction factor (see Table 8) for different buckling class, slenderness ratio and yield stress

$$= \frac{1}{\left[\phi + (\phi^2 - \lambda^2)^{0.5} \right]}$$

λ_{m0} = partial safety factor for material strength.

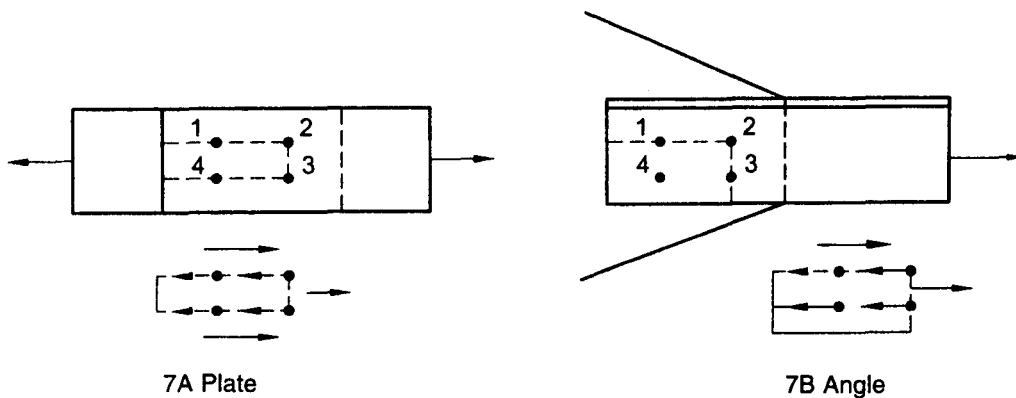


FIG. 7 BLOCK SHEAR FAILURE

NOTE — Calculated values of design compressive stress, f_{cd} for different buckling classes are given in Table 9.

7.1.2.2 The classification of different sections under different buckling class a, b, c or d, is given in Table 10. The stress reduction factor χ , and the design compressive stress f_{cd} , for different buckling class, yield stress, and effective slenderness ratio is given in Table 8 for convenience. The curves corresponding to different buckling class are presented in non-dimensional form, in Fig. 8.

Table 7 Imperfection Factor, α
(Clauses 7.1.1 and 7.1.2.1)

Buckling Class	a	b	c	d
α	0.21	0.34	0.49	0.76

7.2 Effective Length of Compression Members

7.2.1 The effective length KL , is calculated from the actual length L , of the member, considering the rotational and relative translational boundary conditions at the ends. The actual length shall be taken as the length from centre-to-centre of its intersections with the supporting members in the plane of the buckling deformation. In the case of a member with a free end, the free standing length from the center of the intersecting member at the supported end, shall be taken as the actual length.

7.2.2 Effective Length

Where the boundary conditions in the plane of buckling

can be assessed, the effective length, KL can be calculated on the basis of Table 11. Where frame analysis does not consider the equilibrium of a framed structure in the deformed shape (second-order analysis or advanced analysis), the effective length of compression members in such cases can be calculated using the procedure given in D-1. The effective length of stepped column in single storey buildings can be calculated using the procedure given in D-2.

7.2.3 Eccentric Beam Connection

In cases where the beam connections are eccentric in plan with respect to the axes of the column, the same conditions of restraint as in concentric connection shall be deemed to apply, provided the connections are carried across the flange or web of the columns as the case may be, and the web of the beam lies within, or in direct contact with the column section. Where practical difficulties prevent this, the effective length shall be taken as equal to the distance between points of restraint, in non-sway frames.

7.2.4 Compression Members in Trusses

In the case of bolted, riveted or welded trusses and braced frames, the effective length, KL , of the compression members shall be taken as 0.7 to 1.0 times the distance between centres of connections, depending on the degree of end restraint provided. In the case of members of trusses, buckling in the plane perpendicular to the plane of the truss, the effective length, KL shall be taken as the distance between the centres of intersection. The design of angle struts shall be as specified in 7.5.

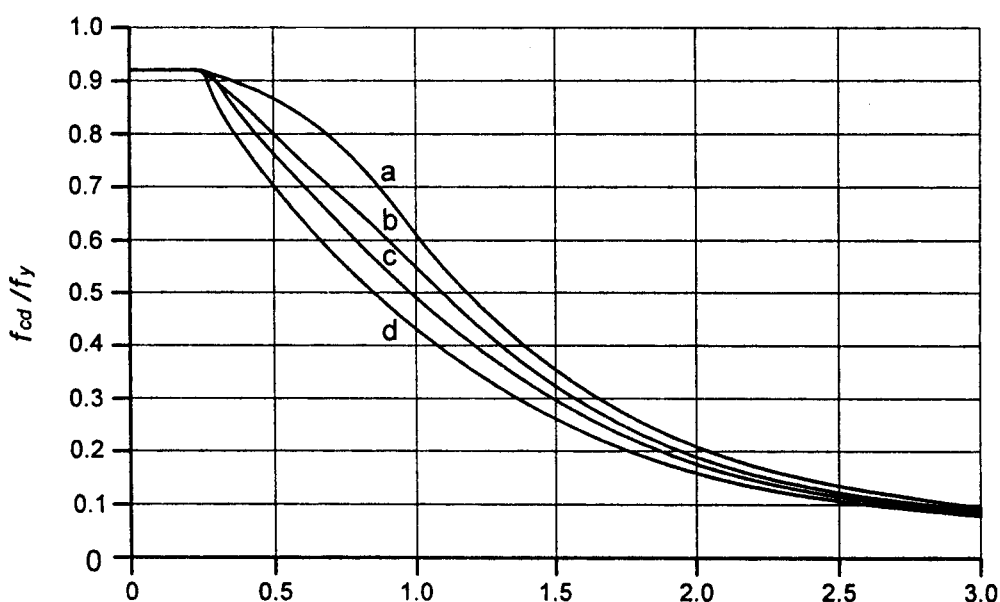


FIG. 8 COLUMN BUCKLING CURVES

Table 8(a) Stress Reduction Factor, χ for Column Buckling Class a
(Clauses 7.1.2.1 and 7.1.2.2)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	0.999	0.998	0.997	0.995	0.994	0.993	0.993	0.990	0.988	0.986	0.984	0.983	0.981	0.979	0.977	0.975	0.972	0.970
30	0.977	0.975	0.974	0.972	0.970	0.969	0.967	0.965	0.961	0.957	0.954	0.951	0.948	0.946	0.943	0.938	0.934	0.930	0.925
40	0.952	0.949	0.947	0.944	0.942	0.939	0.937	0.934	0.926	0.921	0.916	0.911	0.906	0.901	0.896	0.888	0.881	0.873	0.865
50	0.923	0.919	0.915	0.911	0.908	0.904	0.900	0.896	0.884	0.876	0.867	0.859	0.851	0.842	0.834	0.820	0.807	0.794	0.780
60	0.888	0.883	0.877	0.871	0.865	0.859	0.853	0.847	0.828	0.816	0.803	0.790	0.777	0.763	0.750	0.730	0.710	0.690	0.671
70	0.846	0.837	0.829	0.820	0.811	0.803	0.794	0.785	0.758	0.740	0.722	0.703	0.686	0.668	0.651	0.626	0.602	0.579	0.557
80	0.793	0.781	0.769	0.757	0.746	0.734	0.722	0.710	0.675	0.653	0.631	0.610	0.589	0.570	0.551	0.525	0.501	0.478	0.458
90	0.730	0.715	0.700	0.685	0.671	0.657	0.643	0.628	0.590	0.565	0.542	0.520	0.500	0.481	0.463	0.439	0.416	0.396	0.377
100	0.661	0.644	0.627	0.610	0.594	0.579	0.564	0.549	0.510	0.486	0.463	0.443	0.424	0.407	0.390	0.368	0.348	0.331	0.314
110	0.591	0.573	0.555	0.538	0.522	0.507	0.492	0.478	0.440	0.418	0.397	0.379	0.362	0.346	0.332	0.312	0.295	0.279	0.265
120	0.525	0.507	0.489	0.473	0.458	0.443	0.429	0.416	0.381	0.361	0.343	0.326	0.311	0.297	0.284	0.267	0.252	0.238	0.226
130	0.466	0.448	0.432	0.416	0.402	0.388	0.376	0.364	0.332	0.314	0.298	0.283	0.269	0.257	0.246	0.231	0.217	0.206	0.195
140	0.413	0.397	0.382	0.368	0.355	0.342	0.331	0.320	0.291	0.275	0.260	0.247	0.235	0.224	0.214	0.201	0.189	0.179	0.170
150	0.368	0.353	0.339	0.326	0.314	0.303	0.293	0.283	0.257	0.243	0.229	0.218	0.207	0.197	0.189	0.177	0.166	0.157	0.149
160	0.329	0.316	0.303	0.291	0.280	0.270	0.261	0.252	0.229	0.215	0.204	0.193	0.184	0.175	0.167	0.157	0.147	0.139	0.132
170	0.296	0.283	0.272	0.261	0.251	0.242	0.233	0.225	0.204	0.192	0.182	0.172	0.164	0.156	0.149	0.140	0.131	0.124	0.117
180	0.267	0.255	0.245	0.235	0.226	0.218	0.210	0.203	0.184	0.173	0.163	0.155	0.147	0.140	0.134	0.125	0.118	0.111	0.105
190	0.242	0.231	0.222	0.213	0.205	0.197	0.190	0.183	0.166	0.156	0.147	0.140	0.133	0.126	0.121	0.113	0.106	0.100	0.095
200	0.220	0.210	0.202	0.193	0.186	0.179	0.172	0.166	0.151	0.142	0.134	0.127	0.120	0.115	0.109	0.102	0.096	0.091	0.086
210	0.201	0.192	0.184	0.177	0.170	0.163	0.157	0.152	0.137	0.129	0.122	0.115	0.110	0.104	0.099	0.093	0.087	0.083	0.078
220	0.184	0.176	0.169	0.162	0.155	0.149	0.144	0.139	0.126	0.118	0.111	0.106	0.100	0.095	0.091	0.085	0.080	0.075	0.071
230	0.170	0.162	0.155	0.149	0.143	0.137	0.132	0.128	0.115	0.108	0.102	0.097	0.092	0.088	0.083	0.078	0.073	0.069	0.065
240	0.157	0.149	0.143	0.137	0.132	0.127	0.122	0.118	0.106	0.100	0.094	0.089	0.085	0.081	0.077	0.072	0.068	0.064	0.060
250	0.145	0.138	0.132	0.127	0.122	0.117	0.113	0.109	0.098	0.092	0.087	0.082	0.078	0.074	0.071	0.066	0.062	0.059	0.056

Table 8(b) Stress Reduction Factor, χ for Column Buckling Class b
(Clauses 7.1.2.1 and 7.1.2.2)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	0.998	0.996	0.994	0.993	0.991	0.990	0.986	0.983	0.981	0.978	0.975	0.972	0.970	0.967	0.963	0.960	0.956	0.953
30	0.963	0.961	0.958	0.955	0.953	0.950	0.948	0.943	0.938	0.933	0.929	0.924	0.920	0.915	0.911	0.904	0.898	0.892	0.886
40	0.925	0.921	0.917	0.913	0.909	0.906	0.902	0.895	0.887	0.880	0.873	0.866	0.859	0.852	0.845	0.835	0.825	0.815	0.805
50	0.883	0.877	0.872	0.866	0.861	0.855	0.850	0.839	0.829	0.818	0.808	0.798	0.787	0.777	0.767	0.752	0.737	0.722	0.708
60	0.835	0.827	0.820	0.812	0.805	0.798	0.790	0.775	0.761	0.746	0.732	0.718	0.704	0.691	0.677	0.657	0.638	0.620	0.602
70	0.781	0.771	0.761	0.751	0.742	0.732	0.722	0.703	0.685	0.667	0.649	0.632	0.615	0.599	0.584	0.561	0.540	0.520	0.502
80	0.721	0.709	0.697	0.685	0.673	0.661	0.650	0.627	0.606	0.585	0.566	0.547	0.529	0.512	0.496	0.474	0.453	0.434	0.416
90	0.657	0.643	0.629	0.615	0.602	0.589	0.576	0.552	0.530	0.508	0.488	0.470	0.452	0.436	0.421	0.400	0.380	0.363	0.346
100	0.593	0.577	0.562	0.548	0.534	0.520	0.507	0.483	0.461	0.440	0.421	0.403	0.387	0.372	0.358	0.339	0.321	0.306	0.291
110	0.531	0.515	0.500	0.485	0.471	0.458	0.445	0.422	0.401	0.381	0.364	0.348	0.333	0.319	0.306	0.289	0.274	0.260	0.247
120	0.474	0.458	0.443	0.429	0.416	0.403	0.391	0.370	0.350	0.332	0.316	0.301	0.288	0.276	0.265	0.249	0.236	0.223	0.212
130	0.423	0.408	0.394	0.380	0.368	0.356	0.345	0.325	0.307	0.291	0.276	0.263	0.251	0.240	0.230	0.217	0.204	0.194	0.184
140	0.378	0.364	0.350	0.338	0.327	0.316	0.306	0.287	0.271	0.256	0.243	0.231	0.221	0.211	0.202	0.190	0.179	0.169	0.161
150	0.339	0.325	0.313	0.302	0.291	0.281	0.272	0.255	0.241	0.227	0.215	0.205	0.195	0.186	0.178	0.167	0.158	0.149	0.142
160	0.305	0.292	0.281	0.271	0.261	0.252	0.243	0.228	0.215	0.203	0.192	0.182	0.174	0.166	0.158	0.149	0.140	0.133	0.126
170	0.275	0.264	0.253	0.244	0.235	0.227	0.219	0.205	0.193	0.182	0.172	0.163	0.155	0.148	0.142	0.133	0.125	0.118	0.112
180	0.249	0.239	0.229	0.220	0.212	0.205	0.198	0.185	0.174	0.164	0.155	0.147	0.140	0.133	0.128	0.120	0.113	0.106	0.101
190	0.227	0.217	0.208	0.200	0.193	0.186	0.179	0.168	0.157	0.148	0.140	0.133	0.127	0.121	0.115	0.108	0.102	0.096	0.091
200	0.207	0.198	0.190	0.183	0.176	0.169	0.163	0.153	0.143	0.135	0.128	0.121	0.115	0.110	0.105	0.098	0.092	0.087	0.083
210	0.190	0.182	0.174	0.167	0.161	0.155	0.149	0.140	0.131	0.123	0.117	0.110	0.105	0.100	0.096	0.090	0.084	0.080	0.075
220	0.174	0.167	0.160	0.154	0.148	0.142	0.137	0.128	0.120	0.113	0.107	0.101	0.096	0.092	0.088	0.082	0.077	0.073	0.069
230	0.161	0.154	0.147	0.141	0.136	0.131	0.126	0.118	0.111	0.104	0.098	0.093	0.088	0.084	0.080	0.075	0.071	0.067	0.063
240	0.149	0.142	0.136	0.131	0.126	0.121	0.117	0.109	0.102	0.096	0.091	0.086	0.082	0.078	0.074	0.070	0.065	0.062	0.058
250	0.138	0.132	0.126	0.121	0.117	0.112	0.108	0.101	0.095	0.089	0.084	0.080	0.076	0.072	0.069	0.064	0.060	0.057	0.054

Table 8(c) Stress Reduction Factor, χ for Column Buckling Class c
(Clauses 7.1.2.1 and 7.1.2.2)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	0.999	0.997	0.994	0.992	0.990	0.987	0.985	0.981	0.976	0.972	0.968	0.964	0.961	0.957	0.953	0.948	0.943	0.938	0.933
30	0.948	0.944	0.941	0.937	0.933	0.930	0.926	0.920	0.913	0.907	0.901	0.895	0.889	0.883	0.877	0.869	0.861	0.853	0.845
40	0.896	0.891	0.885	0.880	0.875	0.870	0.866	0.856	0.847	0.838	0.829	0.820	0.812	0.803	0.795	0.783	0.771	0.760	0.748
50	0.841	0.834	0.827	0.821	0.814	0.807	0.801	0.788	0.776	0.763	0.752	0.740	0.729	0.717	0.706	0.690	0.675	0.660	0.645
60	0.783	0.774	0.765	0.757	0.748	0.740	0.732	0.716	0.700	0.685	0.670	0.656	0.642	0.628	0.615	0.596	0.578	0.561	0.544
70	0.722	0.711	0.700	0.690	0.680	0.670	0.660	0.641	0.623	0.605	0.588	0.572	0.557	0.542	0.528	0.508	0.489	0.471	0.454
80	0.659	0.646	0.634	0.622	0.611	0.600	0.589	0.568	0.548	0.529	0.512	0.495	0.479	0.464	0.450	0.430	0.412	0.395	0.379
90	0.596	0.583	0.569	0.557	0.544	0.533	0.521	0.499	0.479	0.460	0.443	0.426	0.411	0.397	0.383	0.365	0.348	0.332	0.318
100	0.536	0.522	0.508	0.495	0.483	0.471	0.459	0.438	0.418	0.400	0.384	0.368	0.354	0.341	0.328	0.311	0.296	0.282	0.269
110	0.480	0.466	0.453	0.440	0.428	0.416	0.405	0.385	0.366	0.349	0.333	0.319	0.306	0.294	0.283	0.268	0.254	0.242	0.230
120	0.430	0.416	0.403	0.391	0.379	0.368	0.358	0.339	0.321	0.306	0.291	0.278	0.267	0.256	0.246	0.232	0.220	0.209	0.199
130	0.385	0.372	0.360	0.348	0.337	0.327	0.317	0.299	0.283	0.269	0.256	0.244	0.234	0.224	0.215	0.203	0.192	0.182	0.173
140	0.346	0.333	0.322	0.311	0.301	0.291	0.282	0.266	0.251	0.238	0.227	0.216	0.206	0.197	0.189	0.178	0.168	0.160	0.152
150	0.311	0.300	0.289	0.279	0.269	0.261	0.252	0.237	0.224	0.212	0.202	0.192	0.183	0.175	0.168	0.158	0.149	0.141	0.134
160	0.281	0.270	0.260	0.251	0.242	0.234	0.227	0.213	0.201	0.190	0.180	0.172	0.164	0.156	0.150	0.141	0.133	0.126	0.120
170	0.255	0.245	0.236	0.227	0.219	0.212	0.205	0.192	0.181	0.171	0.162	0.154	0.147	0.140	0.134	0.126	0.119	0.113	0.107
180	0.232	0.223	0.214	0.206	0.199	0.192	0.186	0.174	0.164	0.155	0.147	0.139	0.133	0.127	0.121	0.114	0.107	0.102	0.096
190	0.212	0.203	0.195	0.188	0.181	0.175	0.169	0.158	0.149	0.140	0.133	0.126	0.120	0.115	0.110	0.103	0.097	0.092	0.087
200	0.194	0.186	0.179	0.172	0.166	0.160	0.154	0.144	0.136	0.128	0.121	0.115	0.110	0.105	0.100	0.094	0.089	0.084	0.079
210	0.178	0.171	0.164	0.158	0.152	0.146	0.141	0.132	0.124	0.117	0.111	0.105	0.100	0.096	0.092	0.086	0.081	0.076	0.072
220	0.164	0.157	0.151	0.145	0.140	0.135	0.130	0.122	0.114	0.108	0.102	0.097	0.092	0.088	0.084	0.079	0.074	0.070	0.066
230	0.152	0.145	0.140	0.134	0.129	0.124	0.120	0.112	0.105	0.099	0.094	0.089	0.085	0.081	0.077	0.073	0.068	0.065	0.061
240	0.141	0.135	0.129	0.124	0.120	0.115	0.111	0.104	0.098	0.092	0.087	0.082	0.078	0.075	0.071	0.067	0.063	0.060	0.056
250	0.131	0.125	0.120	0.115	0.111	0.107	0.103	0.096	0.090	0.085	0.081	0.076	0.073	0.069	0.066	0.062	0.058	0.055	0.052

Table 8(d) Stress Reduction Factor, χ for Column Buckling Class d
(Clauses 7.1.2.1 and 7.1.2.2)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	0.999	0.995	0.991	0.988	0.984	0.980	0.977	0.970	0.964	0.958	0.952	0.946	0.940	0.935	0.930	0.922	0.915	0.908	0.901
30	0.922	0.916	0.911	0.906	0.901	0.896	0.891	0.881	0.872	0.863	0.855	0.847	0.839	0.831	0.823	0.813	0.802	0.792	0.782
40	0.848	0.841	0.834	0.828	0.821	0.815	0.808	0.796	0.784	0.773	0.762	0.751	0.741	0.731	0.721	0.707	0.694	0.681	0.668
50	0.777	0.768	0.760	0.752	0.744	0.736	0.728	0.713	0.699	0.685	0.672	0.659	0.647	0.635	0.624	0.608	0.592	0.577	0.563
60	0.707	0.697	0.687	0.678	0.668	0.659	0.651	0.634	0.617	0.602	0.587	0.573	0.560	0.547	0.535	0.517	0.501	0.486	0.471
70	0.640	0.629	0.618	0.607	0.597	0.587	0.578	0.559	0.542	0.526	0.510	0.496	0.482	0.469	0.456	0.439	0.423	0.408	0.394
80	0.576	0.564	0.553	0.542	0.531	0.521	0.511	0.492	0.474	0.458	0.442	0.428	0.414	0.402	0.390	0.373	0.358	0.344	0.330
90	0.517	0.505	0.493	0.482	0.471	0.461	0.451	0.432	0.415	0.399	0.384	0.370	0.357	0.345	0.334	0.319	0.304	0.292	0.280
100	0.464	0.451	0.440	0.428	0.418	0.408	0.398	0.380	0.363	0.348	0.334	0.321	0.309	0.298	0.288	0.274	0.261	0.249	0.239
110	0.416	0.404	0.392	0.381	0.371	0.361	0.352	0.335	0.319	0.305	0.292	0.281	0.270	0.259	0.250	0.237	0.226	0.215	0.206
120	0.373	0.361	0.350	0.340	0.330	0.321	0.313	0.297	0.282	0.269	0.257	0.246	0.236	0.227	0.219	0.207	0.197	0.187	0.179
130	0.336	0.325	0.314	0.305	0.295	0.287	0.279	0.264	0.251	0.239	0.228	0.218	0.209	0.200	0.193	0.182	0.173	0.164	0.157
140	0.303	0.292	0.283	0.274	0.265	0.257	0.250	0.236	0.224	0.213	0.203	0.194	0.185	0.178	0.171	0.161	0.153	0.145	0.138
150	0.274	0.264	0.255	0.247	0.239	0.231	0.224	0.212	0.201	0.190	0.181	0.173	0.165	0.159	0.152	0.144	0.136	0.129	0.123
160	0.249	0.240	0.231	0.223	0.216	0.209	0.203	0.191	0.181	0.171	0.163	0.155	0.149	0.142	0.137	0.129	0.122	0.116	0.110
170	0.227	0.218	0.210	0.203	0.196	0.190	0.184	0.173	0.164	0.155	0.147	0.140	0.134	0.128	0.123	0.116	0.110	0.104	0.099
180	0.207	0.199	0.192	0.185	0.179	0.173	0.167	0.157	0.149	0.141	0.134	0.127	0.122	0.116	0.111	0.105	0.099	0.094	0.089
190	0.190	0.183	0.176	0.169	0.164	0.158	0.153	0.144	0.136	0.128	0.122	0.116	0.111	0.106	0.101	0.095	0.090	0.085	0.081
200	0.175	0.168	0.162	0.156	0.150	0.145	0.140	0.132	0.124	0.118	0.112	0.106	0.101	0.097	0.093	0.087	0.082	0.078	0.074
210	0.161	0.155	0.149	0.143	0.138	0.134	0.129	0.121	0.114	0.108	0.102	0.097	0.093	0.089	0.085	0.080	0.075	0.071	0.068
220	0.149	0.143	0.138	0.133	0.128	0.123	0.119	0.112	0.105	0.100	0.094	0.090	0.086	0.082	0.078	0.074	0.069	0.066	0.062
230	0.138	0.133	0.128	0.123	0.118	0.114	0.110	0.104	0.097	0.092	0.087	0.083	0.079	0.075	0.072	0.068	0.064	0.061	0.058
240	0.129	0.123	0.119	0.114	0.110	0.106	0.103	0.096	0.090	0.085	0.081	0.077	0.073	0.070	0.067	0.063	0.059	0.056	0.053
250	0.120	0.115	0.110	0.106	0.102	0.099	0.095	0.089	0.084	0.079	0.075	0.071	0.068	0.065	0.062	0.058	0.055	0.052	0.049